# End-to-End Machine Learning: Predicting Next-Day Stock Direction Across Four Stocks

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Applied Data Science: Project 4

## Abstract

Predicting short-term stock movement remains a challenging problem because financial markets are characterized by noise, changing macroeconomic conditions, non-stationarity, and substantial limits to predictable structure. This project develops an end-to-end machine learning pipeline to predict whether the adjusted closing price of four large-cap technology stocks, Apple, Netflix, Microsoft, and Amazon, will increase or decrease on the following trading day. Using historical daily market data obtained from Yahoo Finance, the analysis constructs technical and price-based indicators, applies unsupervised learning to identify latent market regimes, and evaluates three supervised classification models: logistic regression, random forest, and XGBoost. To preserve temporal integrity and reduce look-ahead bias, the modeling workflow uses chronological train-test splits and fits unsupervised transformations exclusively on the training data before applying them to the test set. Model performance is assessed through accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). Although ensemble models provide greater flexibility for capturing nonlinear relationships, logistic regression achieves the strongest overall AUC performance across most tickers, suggesting that relatively simple and interpretable models can remain competitive when supported by disciplined feature engineering. The report further emphasizes data quality controls, reproducibility, model-selection rationale, coefficient behavior, feature importance, and a survival-analysis extension for estimating time until large downside events. Finally, an interactive Shiny dashboard is developed to communicate model outputs and allow users to explore results across stocks.

**GitHub Repository:** Project GitHub Repository

**Interactive Shiny Application:** Interactive Shiny Dashboard

## 1. Introduction

Financial markets are highly dynamic systems shaped by both measurable information and unpredictable external forces. Predicting whether a stock price will rise or fall on the following trading day is therefore a widely studied but inherently difficult classification problem. Despite this difficulty, the task provides a valuable framework for examining the full machine learning lifecycle, including data collection, preprocessing, feature engineering, model development, validation, evaluation, interpretation, and communication of results.

The objective of this project is to predict next-day stock direction for four major technology companies: Apple (AAPL), Netflix (NFLX), Microsoft (MSFT), and Amazon (AMZN). These firms were selected because they are highly liquid, heavily traded, and broadly representative of large-cap technology equities, while still exhibiting distinct volatility profiles and market behavior. The project therefore enables both stock-specific modeling and comparative analysis across related but behaviorally different assets.

The prediction target is defined as a binary outcome. A stock is labeled **UP** if its next-day adjusted closing price is higher than its current adjusted closing price. A stock is labeled **DOWN** if its next-day adjusted closing price is lower than or equal to its current adjusted closing price. Rather than focusing exclusively on predictive performance, the project emphasizes model interpretability, temporal validation discipline, and an understanding of how engineered financial indicators contribute to stock movement prediction.

## 2. Data Collection and Preparation

### 2.1 Data Source

The dataset consists of daily stock market data for AAPL, NFLX, MSFT, and AMZN, spanning January 2015 through April 2026. Each stock contains 2,827 daily observations and no missing values in the core raw variables, which supports consistency across the modeling pipeline and reduces the need for extensive imputation. The raw Yahoo Finance variables include standard market indicators such as opening price, daily high and low prices, closing price, adjusted closing price, and trading volume.

Yahoo Finance was used because it provides publicly avail-

able historical market data and is commonly used in financial modeling and academic projects. The consistent date range and standardized column structure across stocks enabled reliable comparative analysis and streamlined subsequent feature engineering, principal component analysis, clustering, and predictive modeling tasks.

### 2.2 Data Quality and Reproducibility

Although the raw OHLCV fields contain no missing values, the dataset is still analytically complex because it is a multi-stock financial time series with temporal dependence, rolling-window feature construction, unequal market regimes, and a prediction target that must be aligned with future outcomes without leaking information. The raw data, processed data objects, modeling scripts, Shiny application code, and supporting documentation are organized in the project repository so that the workflow can be reviewed and reproduced. This structure directly supports reproducibility by separating data acquisition, feature construction, modeling, evaluation, and dashboard deployment.

### 2.3 Data Preparation

Financial time series require careful preprocessing because of temporal dependence and rolling-window calculations. Several preparation steps were applied. Technical indicators were computed using rolling windows, which naturally created missing values at the beginning of each stock series. Rows with incomplete indicator history were removed to ensure that all model inputs were fully observed. Target labels were shifted forward by one trading day so that current-day features were aligned with future stock direction. Finally, the data was split chronologically into training and testing sets, using an 80% training split and a 20% testing split.

Maintaining chronological order is critical in time series modeling. Random shuffling would allow future observations to influence training behavior and would therefore introduce information leakage. The use of a chronological split preserves a more realistic forecasting structure and supports valid out-of-sample evaluation.

Continuous predictors were standardized using parameters estimated from the training data, and the same transformations were then applied to the test data. This training-only preprocessing rule was also used for PCA and $k$-means clustering, ensuring that no information from the held-out test period influenced feature scaling, dimensionality reduction, or unsupervised regime assignment. The resulting processed dataset therefore reflects a forecasting setting in which all transformations are learned from past information and applied to future observations.

## 3. Exploratory Data Analysis

Figure 1 presents the indexed adjusted closing prices for Apple, Netflix, Microsoft, and Amazon from 2015 to 2026. Since each series is normalized to a starting value of 100, the figure allows direct comparison of relative growth rather than absolute share prices. The main pattern is that all four stocks appreciate substantially over the sample period, but they do so with different levels of volatility and trend stability. Netflix reaches the highest indexed peak, rising to slightly above 2,500 before declining sharply, while Amazon also shows large cyclical swings. By contrast, Microsoft and Apple follow steadier upward paths, with Microsoft displaying the smoothest long-run growth pattern among the four stocks.

The same figure also shows that the stocks share common market shocks. Around 2022, all four series decline at roughly the same time, which is consistent with a broad technology-sector selloff rather than an isolated firm-specific event. However, the magnitude of the decline differs by stock. Netflix and Amazon display larger swings, suggesting greater sensitivity to investor sentiment and changing growth expectations. Microsoft shows the most stable long-term trajectory, while Apple combines strong growth with comparatively moderate fluctuations. These visual differences are important because they indicate that the predictive task is not identical across tickers, even though all four firms belong to the large-cap technology universe.

The EDA stage therefore directly guides later modeling decisions. From Figure 1, raw price levels alone would not be sufficient because the stocks differ in volatility, trend persistence, and downside exposure. This motivated the inclusion of momentum indicators, moving-average ratios, volatility measures, and liquidity-related variables. In this way, EDA is not treated as a separate descriptive step only; it provides the empirical motivation for feature engineering, unsupervised regime detection, and model interpretation.

## 4. Feature Engineering

Feature engineering is a central component of the modeling pipeline because it transforms raw stock market variables into predictors that more directly represent market behavior. Instead of relying only on daily price and volume information, the project constructs a broad set of technical indicators designed to capture momentum, trend strength, volatility, relative price position, and trading activity.

To measure momentum, the analysis includes the Relative Strength Index (RSI), Rate of Change (ROC), Williams %R, MACD, and the MACD signal line. These indicators help identify price strength, weakening trends, and potential reversals. To represent trading activity and volume pressure, the project incorporates On-Balance Volume (OBV), which tracks whether trading volume is flowing into or out of a stock. For trend analysis, weekly, quarterly, and annual moving averages are calculated, along with ratios comparing the current price

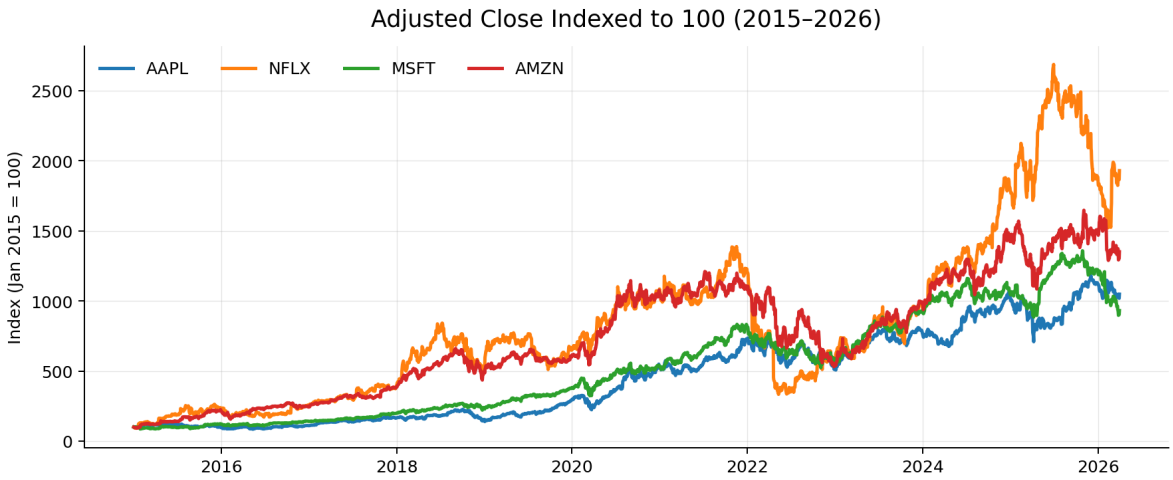



**Figure 1:** Indexed adjusted closing price trends for AAPL, NFLX, MSFT, and AMZN.

to those moving averages. These ratios help measure how far a stock is trading from short- and long-term trend levels.

The feature set also includes intraday ratio variables, including Open-to-Adjusted-Close, High-to-Adjusted-Close, and Low-to-Adjusted-Close. These features capture price movement within the trading day and provide additional information about intraday strength and weakness. Collectively, the engineered features give the models a richer representation of market conditions than raw prices alone and improve their ability to detect patterns associated with next-day price direction.

All engineered variables were constructed using information available at or before the prediction date. This design is important because technical indicators based on future observations would artificially inflate performance and undermine the validity of the forecasting task. By combining current-day ratios, lagged returns, rolling measures, and training-only transformations, the feature engineering process balances predictive richness with methodological discipline.

## 5. Unsupervised Learning Integration

To capture hidden market structure, unsupervised learning is incorporated into the modeling pipeline. The purpose of this step is not to predict the target directly, but rather to identify latent regimes in the feature space that may provide useful contextual information for supervised models.

### 5.1 Principal Component Analysis

Figure 2 shows the proportion of variance explained by each principal component for AAPL-related features. The first component explains 44.5% of the total variance, the second explains 27.0%, and the third explains 13.4%. Together, the first three components account for approximately 84.9% of the total variance, meaning that most of the information in the original feature set can be summarized using a much smaller

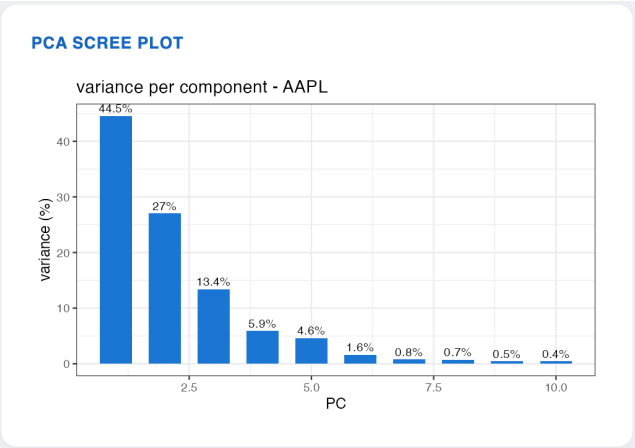
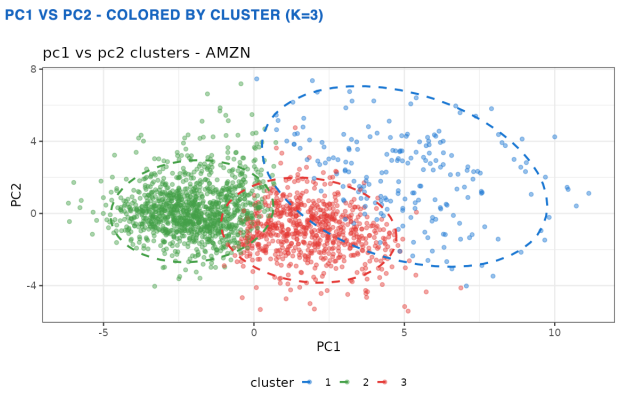


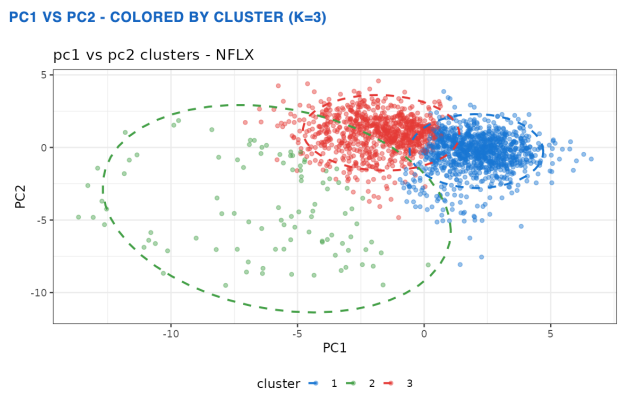

**Figure 2:** PCA scree plot showing the proportion of variance explained by each principal component.

number of dimensions. The fourth and fifth components explain 5.9% and 4.6%, respectively, so retaining the first five components preserves approximately 95.4% of the total variance. The sharp drop after the third component suggests diminishing marginal returns from retaining many additional components.

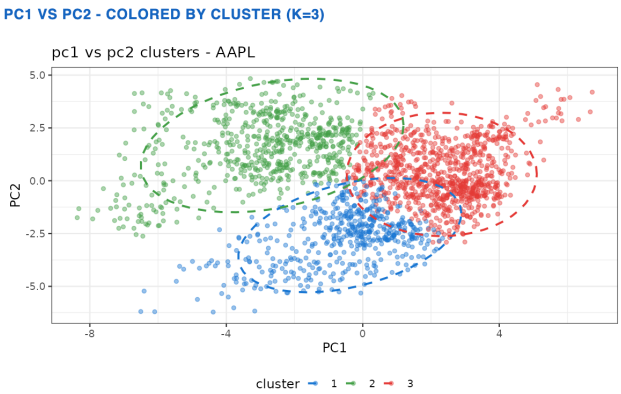
After reducing the dimensionality of the feature space using PCA, the first five principal components are retained to preserve most of the dataset's variance while removing redundancy among highly correlated variables. These components provide a compact representation of underlying market behavior and serve as inputs for $k$-means clustering. By projecting the data into a lower-dimensional space, clustering can more effectively identify latent market regimes and stock behavior patterns that may not be visible in the original high-dimensional feature set.



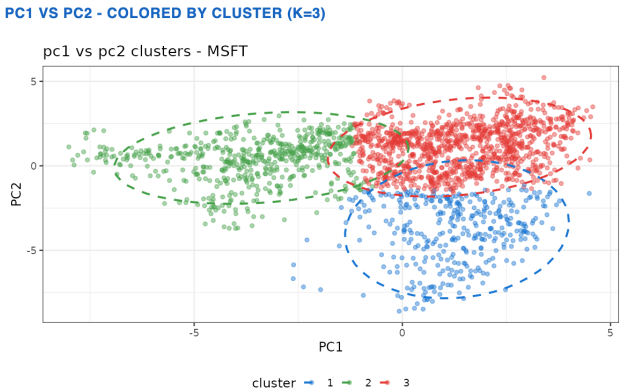

**(a)** AMZN.




**(b)** NFLX.




**(c)** AAPL.




**(d)** MSFT.

**Figure 3:** K-means clustering results with $k = 3$ using PCA-transformed features across the four stocks.

### 5.2 K-Means Clustering

Figure 3 visualizes the $k$-means clustering results with $k = 3$ in the PCA-transformed feature space for all four stocks. The cluster separation is not identical across tickers, which suggests that latent market regimes differ by stock. In the Netflix panel, the blue and red clusters are relatively concentrated on the right side of the first principal component, while the green cluster is more dispersed toward the left. AMZN also shows a relatively wide blue cluster on the positive-PC1 side and more overlap between the green and red clusters near the center-left region. AAPL and MSFT exhibit three recognizable regimes, but the clusters overlap more around the central region of the PCA space. These cluster labels are then added as categorical features for supervised learning, allowing the supervised models to account for higher-level market states, such as trending, volatile, or consolidating periods, rather than relying exclusively on raw technical indicators.

## 6. Supervised Modeling

For the supervised modeling stage, three classification models are trained and compared: logistic regression, random forest, and XGBoost. Logistic regression is used as the baseline model because of its simplicity, computational efficiency, and interpretability. Its coefficients make it possible to examine how each feature influences the probability of upward or downward price movement. However, its main limitation is the assumption of an approximately linear relationship between the input features and the target outcome, which may not fully capture the complexity of financial market behavior.

Random forest is used as a more flexible alternative. As an ensemble of decision trees, it can model nonlinear relationships and feature interactions while remaining relatively robust to overfitting. However, it is less directly interpretable than logistic regression. XGBoost is also included because gradient boosting methods are known for strong predictive performance on structured data and for their ability to model complex nonlinear patterns. At the same time, XGBoost is more computationally intensive and generally requires more careful hyperparameter tuning.

### 6.1 Validation and Tuning Strategy

Model development follows a time-aware validation structure. The models are trained on the earlier portion of each stock's history and evaluated on the later held-out period, rather than using random train-test splits. This approach is

more appropriate for financial forecasting because it mirrors the real-world sequence in which past data are used to predict future observations. Model comparison is based on out-of-sample performance rather than in-sample fit, reducing the risk of over-interpreting patterns that may not generalize.

The three algorithms were selected to represent a range of complexity: a transparent linear baseline, a bagging-based nonlinear ensemble, and a boosting-based nonlinear ensemble. This design supports a meaningful comparison between interpretability, flexibility, and predictive performance. Any model selection decision is therefore evaluated not only through metrics, but also through robustness, ease of explanation, and suitability for a noisy financial prediction problem.

## 7. Model Evaluation

Figure 4 summarizes the quantitative performance of each model by ticker. The table shows that logistic regression achieves the highest AUC for AAPL, NFLX, MSFT, and AMZN, with AUC values around 0.94 for AAPL, 0.97 for NFLX, 0.92 for MSFT, and 0.95 for AMZN. Random forest and XGBoost remain competitive, but they generally do not exceed logistic regression on the main discrimination metric. This is an important finding because it shows that greater model complexity does not necessarily translate into better out-of-sample performance even in a noisy financial prediction task.

Figure 5 provides the visual counterpart to the metric table. Across the four ROC panels, all three models remain above the diagonal baseline, indicating that they perform better than random classification. The logistic regression curve is generally closest to the upper-left corner, with the largest visual separation from the ensemble models appearing for AAPL and MSFT. For NFLX and AMZN, the three curves are closer together, but logistic regression still has the highest AUC in the reported metrics. Given next-day stock direction is often difficult to predict using raw accuracy alone, AUC is particularly useful here: it evaluates the model's ability to rank positive cases above negative cases across many possible thresholds rather than depending on one arbitrary cutoff.

### 7.1 Model Selection Rationale

Based on the combined evidence from Figure 4 and Figure 5, logistic regression is the most defensible final model for this project. It provides the strongest and most consistent AUC performance across the tickers while also offering direct interpretability through coefficient estimates and odds ratios. Although random forest and XGBoost are more flexible, their additional complexity does not produce a clear performance advantage in this setting. Given the weak-signal nature of next-day stock prediction, the final model selection prioritizes out-of-sample discrimination, transparency, and robustness over model complexity alone.

## 8. Model Interpretation

Interpretability is analyzed differently across the models in order to understand how each approach captures relationships within the financial data. For logistic regression, interpretability focuses on examining model coefficients and feature-log odds relationships, allowing direct analysis of how individual features influence the probability of upward or downward price movement. Random forest interpretability is examined through feature importance rankings, which summarize the relative contribution of predictors to model outputs. XGBoost is interpreted using feature importance measures that help identify which variables are most influential in its boosted decision-tree structure. Together, these methods provide insight into both linear and nonlinear decision patterns.

### 8.1 Logistic Regression Coefficients

Figure 6 reports the logistic regression odds ratios for AAPL with 95% confidence intervals. Values greater than 1 indicate features associated with higher odds of an upward next-day move, while values below 1 indicate features associated with lower odds of an upward move. In the figure, PriceROC and HighCloseRatio have the largest odds ratios, and RSI is also positioned above 1, suggesting that momentum and relative intraday price strength are important bullish signals in the logistic regression model. The cluster indicators also appear on the positive side, although their wider confidence intervals indicate greater uncertainty than the strongest price-based signals.

Figure 6 also shows that several variables fall below 1, including OBV, MACD, OpenCloseRatio, and multiple moving-average ratio features. These negative-side estimates indicate lower odds of an upward next-day move in this specification. The confidence intervals are useful because they show uncertainty around each estimated effect: wider intervals indicate less stable coefficient estimates, while narrower intervals suggest more reliable relationships. Overall, the coefficient figure connects the model's predictive performance to interpretable financial mechanisms: momentum and relative price-strength variables increase predicted upside probability, whereas several trend-ratio and volume-related signals reduce it.

### 8.2 Feature Importance

Figure 7 and Figure 8 identify the variables that contribute most strongly to predicting AAPL stock movement in the two ensemble models. In both figures, WeeklyMeanRatio and OpenCloseRatio are the two most important predictors, suggesting that short-term trend position and the relationship between opening and closing prices are especially informative. The random forest plot also ranks LowCloseRatio, stochastic oscillator measures, RSI, and HighCloseRatio highly. The XGBoost plot similarly emphasizes StoFastD, StoSlowD, LowCloseRatio, and StoFastK after the top two features. This reinforces the broader conclusion that trend-position, intraday

| ticker | model | accuracy | sensitivity | specificity | precision | recall | f1 | auc |
|---|---|---|---|---|---|---|---|---|
| AAPL | Logistic Regression | 0.8718 | 0.8702 | 0.8739 | 0.8953 | 0.8702 | 0.8826 | 0.9409 |
| AAPL | Random Forest | 0.7456 | 0.9684 | 0.4696 | 0.6935 | 0.9684 | 0.8082 | 0.8518 |
| AAPL | XGBoost | 0.7806 | 0.9404 | 0.5826 | 0.7363 | 0.9404 | 0.8259 | 0.8684 |
| NFLX | Logistic Regression | 0.8971 | 0.9323 | 0.8594 | 0.8763 | 0.9323 | 0.9035 | 0.9666 |
| NFLX | Random Forest | 0.8602 | 0.9060 | 0.8112 | 0.8368 | 0.9060 | 0.8700 | 0.9508 |
| NFLX | XGBoost | 0.8718 | 0.9023 | 0.8394 | 0.8571 | 0.9023 | 0.8791 | 0.9515 |
| MSFT | Logistic Regression | 0.8311 | 0.8376 | 0.8238 | 0.8407 | 0.8376 | 0.8392 | 0.9228 |
| MSFT | Random Forest | 0.6641 | 0.9742 | 0.3197 | 0.6140 | 0.9742 | 0.7532 | 0.8525 |
| MSFT | XGBoost | 0.7592 | 0.9410 | 0.5574 | 0.7025 | 0.9410 | 0.8044 | 0.8818 |
| AMZN | Logistic Regression | 0.8602 | 0.8450 | 0.8770 | 0.8842 | 0.8450 | 0.8642 | 0.9453 |
| AMZN | Random Forest | 0.8427 | 0.8524 | 0.8320 | 0.8493 | 0.8524 | 0.8508 | 0.9214 |
| AMZN | XGBoost | 0.8563 | 0.8450 | 0.8689 | 0.8774 | 0.8450 | 0.8609 | 0.9348 |

**Figure 4:** Classification performance metrics by model and ticker.

price-ratio, and momentum-related variables carry much of the predictive signal.

The two feature-importance figures are similar but not identical. Random forest importance in Figure 7 reflects average impurity reduction across many trees, while XGBoost importance in Figure 8 reflects the contribution of variables within a sequential boosting process. As a result, the most influential variables overlap, but the exact ranking of lower-impact predictors differs. This comparison is useful because it shows that the main signals are not entirely model-specific: multiple algorithms point toward trend-position and short-term price-ratio variables as important predictors.

## 9. Extension: Survival Analysis for Time Until Large Downside Move

In addition to daily stock-direction classification, the project includes a survival analysis extension to study downside risk timing. Rather than predicting whether a stock moves upward or downward on a given day, this extension estimates the number of trading days until the stock experiences a large negative-return event. The event is defined as a daily return of $-3\%$ or lower, corresponding to a decrease of at least 3%.

To construct the survival dataset, the daily stock panel is converted into survival spells. Each spell begins on a trading day and ends when the next $-3\%$ daily return event occurs. The duration variable is the number of trading days until that event. If no such event occurs before the end of the sample period, the spell is treated as right-censored. In this context, right censoring means that the event is not observed during the sample window, although it may occur after the sample ends.

A Cox proportional hazards model is fit using technical indicators as covariates, including RSI, MACD, PriceROC, OBV, rolling volatility, and moving-average ratios. The Cox model estimates the hazard of a large downside event as

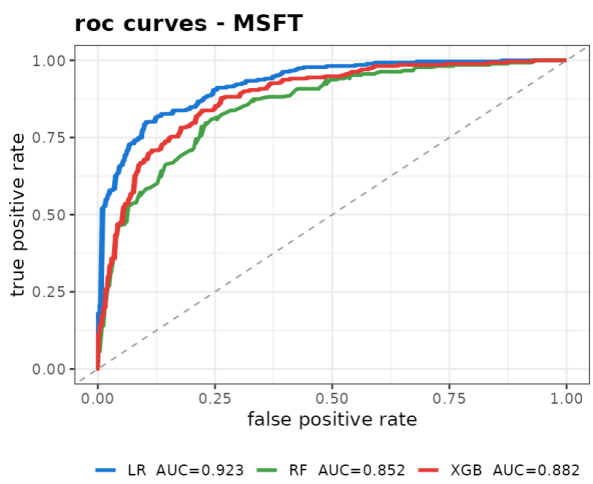
$$h(t \mid X) = h_0(t) \exp(\beta^\top X), \tag{1}$$

where $h(t \mid X)$ is the hazard of a large negative-return event at time $t$, $h_0(t)$ is the baseline hazard, and $X$ represents the vector of technical indicators. A hazard ratio above 1 indicates that higher values of a feature are associated with faster occurrence of a large downside move, while a hazard ratio below 1 indicates lower downside-event risk.

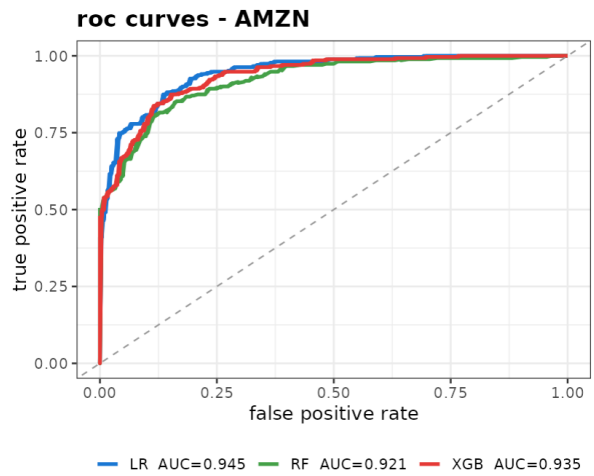
The baseline Cox model achieves a concordance of 0.65, indicating moderate ability to rank observations by downside-event risk. The strongest significant predictors are 21-day rolling volatility and WeeklyMeanRatio. The hazard ratio for 21-day rolling volatility is approximately 1.21, implying that a one-standard-deviation increase in recent volatility is associated with roughly a 21% higher hazard of a $-3\%$ daily return event. WeeklyMeanRatio has a hazard ratio of approximately 1.15, suggesting that short-term price weakness relative to the weekly moving average is also associated with faster downside-event timing.

Figure 9 displays the Cox model hazard ratios with 95% confidence intervals. The vertical reference line at 1 represents no change in downside-event hazard. Variables whose confidence intervals lie mostly to the right of 1 are associated with faster arrival of a $-3\%$ daily return event. In this figure, RollingVolatility_21 and WeeklyMeanRatio are the clearest significant predictors, while most other indicators cross or remain close to the reference line. PriceROC is positioned below 1, indicating that stronger recent price change is associated with lower downside-event hazard in this survival specification. This visual evidence supports the interpretation that recent volatility and short-term price position relative to the weekly mean are the strongest signals of elevated downside-event risk.

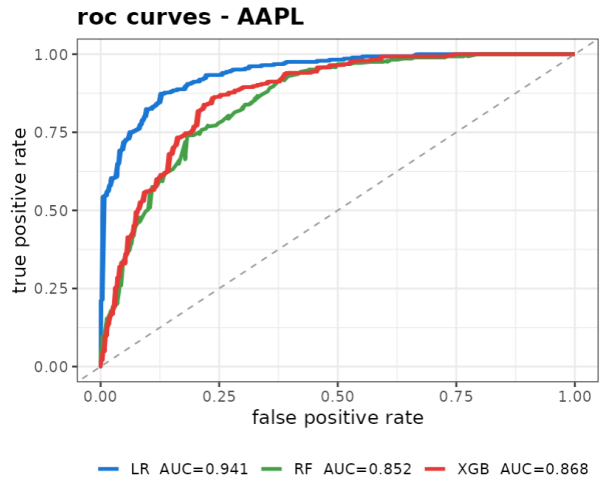
Figure 10 groups spells into low-, medium-, and high-risk terciles based on the Cox model's predicted hazard scores. The Kaplan-Meier curves estimate the probability that a spell has not yet experienced a $-3\%$ daily return event after a given number of trading days. The figure shows clear separation



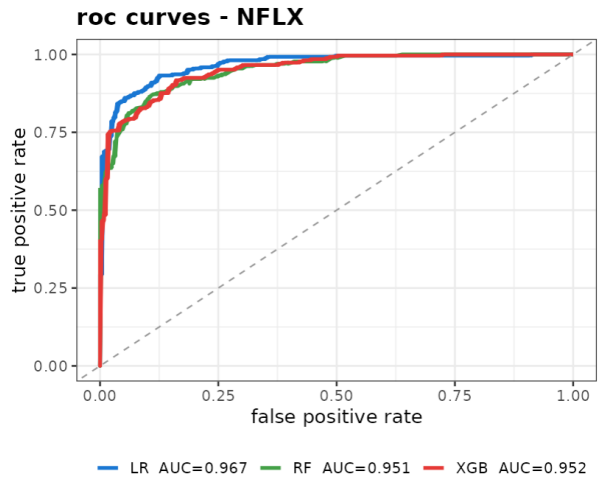

**(a)** MSFT




**(b)** AMZN




**(c)** AAPL




**(d)** NFLX

**Figure 5:** ROC curves for logistic regression, random forest, and XGBoost across the four tickers.

among the three curves: the high-risk group declines most rapidly, the medium-risk group declines at an intermediate rate, and the low-risk group survives the longest. The fitted survival output gives a median time to a $-3\%$ event of 16 trading days for the low-risk group, 8 trading days for the medium-risk group, and 4 trading days for the high-risk group. This ordering is consistent with the visual pattern in Figure 10, where the high-risk curve falls below the others earliest. Together with the significant log-rank test, this separation confirms that the predicted risk groups differ meaningfully in time-to-event behavior.

Overall, the survival analysis provides a risk-oriented complement to the main classification models. While the supervised models predict daily direction, the Cox model estimates how technical indicators relate to the timing of large downside moves. The results suggest that recent volatility and short-term price weakness are important signals for identifying periods of elevated downside-event risk.

## 10. Limitations

Despite the development of a complete modeling pipeline, several important limitations remain. First, next-day stock direction prediction is inherently difficult because financial markets contain weak and unstable signals. Much of daily price movement is driven by noise, unexpected information, and market-wide shocks, leaving only a limited amount of usable predictive structure.

It is also important to note that financial markets are non-stationary. The relationships between technical indicators and future returns may shift over time as macroeconomic conditions, investor behavior, market liquidity, and firm-specific expectations evolve. As a result, model performance observed during one period may not necessarily generalize to future market conditions.

The analysis predicts direction but does not incorporate transaction costs, bid-ask spreads, liquidity constraints, or slippage.

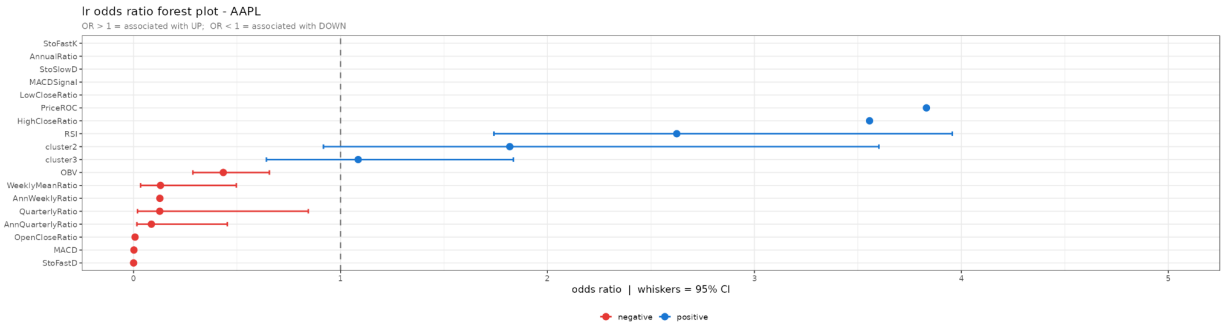



**Figure 6:** Logistic regression odds-ratio coefficient plot with confidence intervals for AAPL.

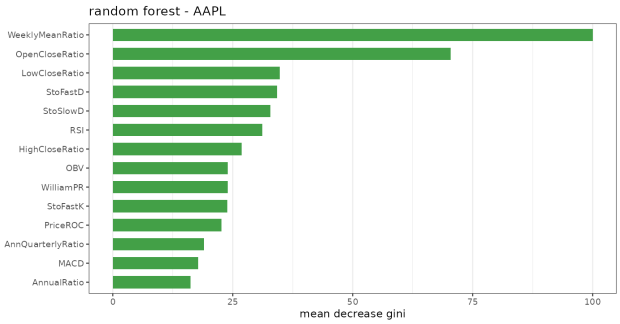



**Figure 7:** Random forest feature importance rankings for AAPL.

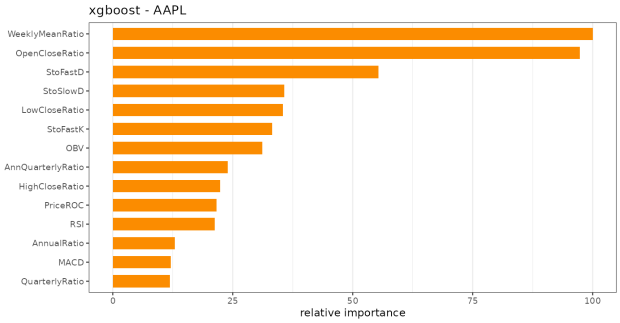



**Figure 8:** XGBoost feature importance rankings for AAPL.

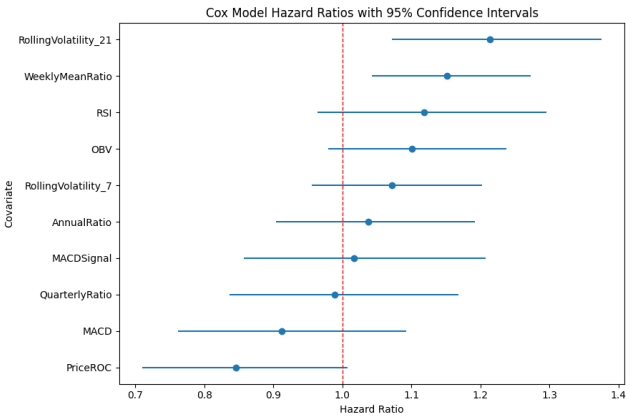



**Figure 9:** Cox proportional hazards model hazard ratio forest plot with 95% confidence intervals.

These factors can materially affect real-world trading performance and may cause a statistically accurate prediction model to be economically unprofitable. Finally, the project is limited to four technology stocks, restricting the generalizability of the findings. A broader universe of assets would be required to determine whether the observed patterns hold across sectors, market capitalizations, and asset classes.

## 11. Future Work

Several extensions could strengthen the project. For example, walk-forward validation could be used instead of a single chronological train-test split. This would better approximate real trading environments, where models are periodically retrained and evaluated as new data becomes available. Also, the model could be expanded to include broader macroeconomic variables, such as interest rates, inflation indicators, market-wide volatility, and sector-level returns. These variables may capture external conditions that technical indicators alone cannot fully represent.

News sentiment analysis could be incorporated because market reactions are often driven by qualitative information, including earnings reports, analyst revisions, firm announcements, and macroeconomic news. Probability calibration could improve the reliability of model confidence estimates, which would be especially useful for risk management and decision-making. Finally, reinforcement learning could be explored as a method for moving beyond direction prediction toward the direct optimization of trading strategies over time. Such an approach would allow the model to learn from sequential decisions and market feedback, although it would also require careful attention to transaction costs and realistic trading constraints.

## 12. Reproducibility and Communication

The project is accompanied by the GitHub repository listed on the first page, which contains the code, data objects, and application files needed to reproduce the major components of the analysis. The workflow is organized so that users can trace the pipeline from data collection and feature construction to PCA, clustering, supervised modeling, evaluation, and Shiny

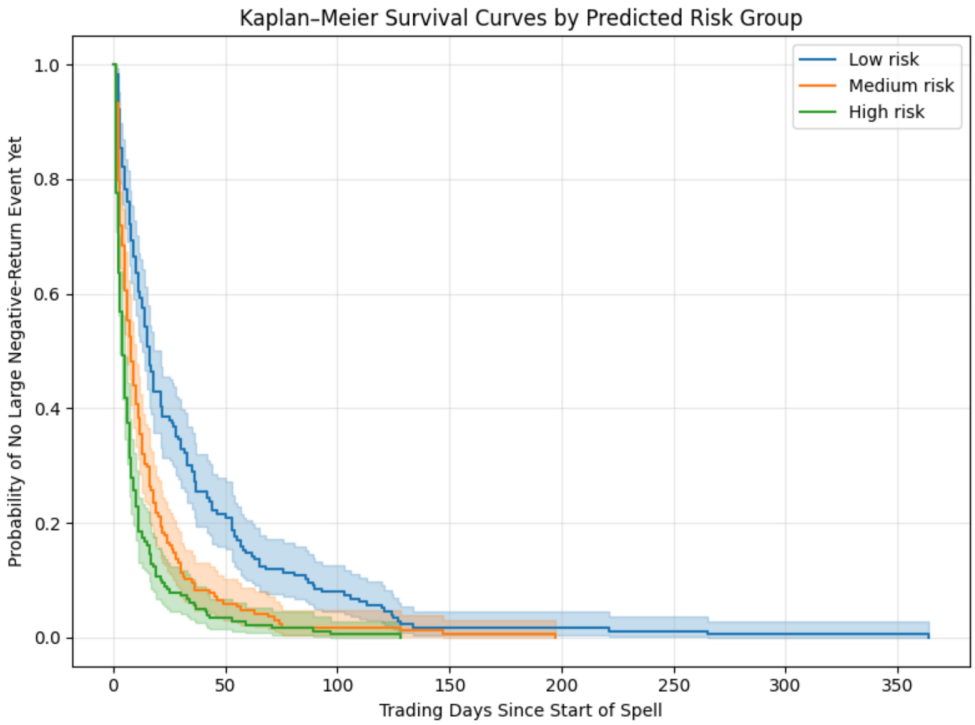



**Figure 10:** Kaplan-Meier survival curves for low-, medium-, and high-risk groups based on predicted Cox model hazard scores.

deployment. The Shiny application listed on the first page serves as an interactive communication layer for the project. Rather than presenting only static model outputs, the dashboard allows users to explore stock-level results, compare model behavior, and examine key visualizations dynamically.

## 13. Conclusion

This project demonstrates the full machine learning lifecycle in the context of financial time series data. Although predicting next-day stock direction remains difficult, the results show that machine learning models can capture weak predictive signals from carefully engineered technical indicators. The comparison across logistic regression, random forest, and XGBoost highlights the tradeoff between interpretability and model flexibility. In this project, logistic regression performs particularly well in terms of AUC, suggesting that simpler models can remain competitive when features are thoughtfully constructed and validation is properly designed.

The inclusion of unsupervised market regime detection, illustrated in Figure 3, adds modeling depth and helps contextualize price behavior. The survival analysis extension, summarized through Figure 9 and Figure 10, further broadens the project by shifting attention from daily direction to downside risk timing. The central conclusion is not that stock prices are easy to predict, but rather that robust predictive systems require both technical rigor and domain awareness.

## Team Member Contributions

The team contributions were distributed as follows:

- **Nikhil Shanbhag:** Survival analysis and feature importance analysis.
- **Charlene Shen:** Unsupervised and supervised model development, and full Shiny application development.
- **Shreya Amalapurapu:** Research report.
- **Yifan Wang:** Presentation slides and Shiny application deployment.